Where does uncertainty actually matter?

Habnetic estimates posterior decision stability to identify the small subset of assets whose prioritisation is actually uncertain.

CITYWIDE OVERVIEW – ROTTERDAM

Habnetic
Resilient Housing Bayes

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1.0 -> high probability of belonging to the top set

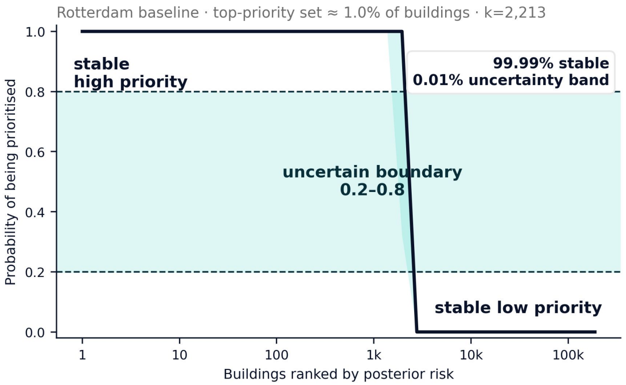
0.5->the posterior cannot confidently decide

0.0->high confidence that it is outside the top set

WHERE SHOULD UNCERTAINTY ANALYSIS FOCUS?

Habnetic estimates **posterior decision stability** for every asset.
Most buildings remain consistently prioritised or consistently excluded across posterior simulations.
Only a small decision boundary contains assets whose prioritisation may change under uncertainty.
In Rotterdam, more than 99% of buildings lie outside this unstable boundary region.

DECISION STABILITY STRUCTURE



ZOOMED EXAMPLE ROTTERDAM

A. Consistently high-priority area

Most assets have very low or very high probability → stable decisions.

B. Boundary area (mixed uncertainty)

Uncertainty concentrates near the prioritisation boundary.

CROSS CITY TRANSFER

Same model, different cities

MODEL FRAMEWORK

Exposure → Hazard → Impact → Posterior ranking

Prioritisation becomes probabilistic instead of deterministic.

WHAT THIS ENABLES

Identify unstable intervention decisions

Focus review on boundary cases

Prioritise robust decisions

DECISION IMPLICATION

Focus review on the uncertain decision boundary

Paper & data
habnetic.org

github.com/habnetic/resilient-housing-bayes

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